

Year 7 · Chapter 4 · Fractions and percentages: answers (teacher copy)

4.01 Fractions (booklet p. 3)

WE DO examples

- Ex 1: $\frac{7}{4}$
- Ex 2: $1\frac{3}{4}$
- Ex 3: $\frac{5}{15}$
- Ex 4: $\frac{12}{16}$
- Ex 5: $\frac{5}{6}$
- Ex 6: $\frac{3}{2} = 1\frac{1}{2}$

Set A

- $\frac{7}{4}$
- $\frac{22}{9}$
- $\frac{28}{5}$
- $\frac{22}{7}$
- $\frac{9}{2}$
- $\frac{17}{6}$

Set B

- $1\frac{3}{4}$
- $8\frac{2}{3}$
- $3\frac{2}{7}$
- $5\frac{1}{2}$
- $3\frac{2}{5}$
- $1\frac{1}{2}$

Set C

- 5
- 16
- 16
- 36
- 25
- 15

Set D

- $\frac{5}{6}$
- $\frac{3}{4}$
- $\frac{3}{4}$
- $\frac{4}{3}$
- $\frac{1}{2}$
- $\frac{4}{7}$

Extension

- $\frac{15}{30}$
- $\frac{10}{35}$

Exit ticket

- Level 1: $\frac{17}{5}$
- Level 2: 10
- Level 3: $\frac{3}{4}$

4.02 Ordering fractions (booklet p. 7)

WE DO examples

- Ex 1: $\frac{5}{8}$
- Ex 2: $\frac{2}{3}$
- Ex 3: $\frac{3}{10}, \frac{2}{5}, \frac{7}{10}, \frac{4}{5}$
- Ex 4: $\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{11}{12}$
- Ex 5: $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
- Ex 6: $\frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$

Set A

- $\frac{5}{7}$
- $\frac{1}{3}$
- $\frac{9}{10}$
- $\frac{1}{2}$
- $\frac{5}{6}$
- $\frac{3}{4}$

Set B

- $\frac{6}{12}$
- $\frac{8}{12}$
- $\frac{9}{12}$
- $\frac{4}{10}$
- $\frac{10}{12}$
- $\frac{9}{24}$

Set C

- $\frac{3}{10}, \frac{2}{5}, \frac{7}{10}, \frac{4}{5}$
- $\frac{1}{3}, \frac{1}{2}, \frac{7}{12}, \frac{11}{12}$
- $\frac{3}{8}, \frac{1}{2}, \frac{5}{8}, \frac{3}{4}$
- $\frac{4}{9}, \frac{1}{2}, \frac{2}{3}, \frac{5}{6}$

Set D

- $\frac{7}{12}, \frac{2}{3}, \frac{3}{4}, \frac{5}{6}$
- $\frac{4}{15}, \frac{1}{2}, \frac{3}{5}, \frac{7}{10}$
- $1\frac{1}{6}, \frac{5}{4}, 1\frac{1}{3}, \frac{11}{8}$
- e.g. $\frac{5}{12}$

Extension

- $\frac{3}{7}, \frac{4}{9}, \frac{1}{2}$
- $\frac{7}{24}$

Exit ticket

- Level 1: $\frac{7}{9}$
- Level 2: $\frac{4}{5}$
- Level 3: $\frac{2}{3}$, $\frac{3}{4}$, $\frac{5}{6}$

4.03 Adding and subtracting fractions

(booklet p. 11)

WE DO examples

- Ex 1: 1
- Ex 2: $\frac{3}{7}$
- Ex 3: $\frac{7}{10}$
- Ex 4: $\frac{11}{20}$
- Ex 5: $\frac{5}{6}$
- Ex 6: $1\frac{7}{12}$

Set A

- 1
- $\frac{7}{9}$
- $\frac{2}{5}$
- $\frac{1}{2}$
- $1\frac{2}{5}$
- $\frac{1}{2}$

Set B

- $\frac{3}{4}$
- $\frac{1}{2}$
- $\frac{7}{10}$
- $\frac{1}{6}$
- $\frac{1}{8}$
- $\frac{7}{8}$

Set C

- $\frac{5}{6}$
- $\frac{11}{20}$
- $\frac{11}{12}$
- $\frac{3}{10}$
- $1\frac{7}{12}$
- $\frac{25}{72}$

Set D

- $1\frac{19}{30}$
- $1\frac{1}{12}$
- $\frac{3}{8}$
- $\frac{4}{15}$

Extension

- $\frac{11}{15}$
- $\frac{1}{3} + \frac{1}{4}$

Exit ticket

- Level 1: $\frac{5}{7}$
- Level 2: $\frac{1}{3}$
- Level 3: $1\frac{4}{15}$

4.04 Adding and subtracting mixed numerals (booklet p. 15)**WE DO examples**

- Ex 1: $8\frac{7}{10}$
- Ex 2: $5\frac{1}{4}$
- Ex 3: $9\frac{2}{3}$
- Ex 4: $1\frac{1}{2}$
- Ex 5: $1\frac{7}{12}$
- Ex 6: $2\frac{13}{20}$

Set A

- $3\frac{3}{5}$
- $5\frac{1}{2}$
- $3\frac{3}{7}$
- $8\frac{1}{2}$
- $4\frac{1}{3}$
- 3

Set B

- $9\frac{2}{3}$
- $4\frac{1}{4}$
- $4\frac{1}{2}$
- $2\frac{3}{5}$
- 4
- $1\frac{3}{4}$

Set C

- $8\frac{7}{10}$
- $1\frac{1}{2}$
- $5\frac{1}{4}$
- $2\frac{5}{12}$
- $4\frac{5}{12}$
- $2\frac{7}{12}$

Set D

- $1\frac{7}{12}$
- $1\frac{7}{10}$
- $6\frac{7}{12}$
- $1\frac{3}{4}$ m

Extension

- $1\frac{1}{12}$
- $3\frac{1}{2}$

Exit ticket

1. Level 1: $6\frac{2}{3}$
2. Level 2: $5\frac{3}{5}$
3. Level 3: $1\frac{7}{12}$

4.05 Fraction of a quantity (booklet p. 19)

WE DO examples

1. Ex 1: 7
2. Ex 2: 9
3. Ex 3: 16 h
4. Ex 4: 1600 m
5. Ex 5: 135 min
6. Ex 6: 50 s

Set A

1. 7
2. 3.5
3. 8
4. 12
5. 20
6. 9

Set B

1. 24
2. 35
3. 105
4. 750
5. 12
6. 40

Set C

1. 16 h
2. 1600 m
3. 135 min
4. 105 mm
5. 50 s
6. 125 cm

Set D

1. 12 boys
2. \$240
3. 36
4. $\frac{3}{4}$ h by 3 min

Extension

1. 33.75
2. 160 L

Exit ticket

1. Level 1: 9
2. Level 2: 60
3. Level 3: 90 min

4.06 Multiplying fractions (booklet p. 23)

WE DO examples

1. Ex 1: $\frac{2}{15}$
2. Ex 2: 6
3. Ex 3: $\frac{5}{14}$
4. Ex 4: $\frac{1}{40}$
5. Ex 5: $\frac{3}{4}$
6. Ex 6: $10\frac{8}{9}$

Set A

1. $\frac{2}{15}$
2. $\frac{1}{21}$
3. $\frac{3}{8}$
4. $\frac{4}{15}$
5. 6
6. 10

Set B

1. $\frac{5}{14}$
2. $\frac{1}{6}$
3. $\frac{1}{2}$
4. $\frac{1}{40}$
5. $\frac{1}{6}$
6. $\frac{3}{5}$

Set C

1. $\frac{3}{4}$
2. $1\frac{1}{10}$
3. $2\frac{1}{4}$
4. $1\frac{1}{3}$
5. $1\frac{3}{4}$
6. $1\frac{7}{20}$

Set D

1. $10\frac{8}{9}$
2. $2\frac{1}{3}$
3. $1\frac{1}{6}$
4. $1\frac{5}{6}$ cups

Extension

1. $\frac{1}{10}$
2. $9\frac{1}{3}$ m²

Exit ticket

1. Level 1: $\frac{3}{20}$
2. Level 2: $\frac{1}{6}$
3. Level 3: $3\frac{1}{3}$

4.07 Dividing fractions (booklet p. 27)

WE DO examples

1. Ex 1: $\frac{4}{3}$, $\frac{1}{5}$, $\frac{2}{5}$
2. Ex 2: 50
3. Ex 3: $\frac{1}{6}$
4. Ex 4: 2
5. Ex 5: 2
6. Ex 6: $1\frac{1}{2}$

Set A

1. $\frac{4}{3}$
2. $\frac{5}{2}$
3. $\frac{1}{7}$
4. 9
5. $\frac{3}{8}$
6. $\frac{2}{3}$

Set B

1. 50
2. $\frac{1}{6}$
3. $1\frac{1}{2}$
4. 9
5. $\frac{2}{5}$
6. 12

Set C

1. 2
2. 6
3. $\frac{32}{35}$
4. $1\frac{5}{12}$
5. $\frac{4}{5}$
6. $\frac{5}{6}$

Set D

1. 3
2. 2
3. $1\frac{1}{2}$
4. 10 pieces

Extension

1. $\frac{2}{3}$
2. 5

Exit ticket

1. Level 1: $\frac{8}{5}$
2. Level 2: 4
3. Level 3: 2

4.08 Fraction problems (booklet p. 31)**WE DO examples**

1. Ex 1: $6\frac{7}{12}$ L
2. Ex 2: $1\frac{1}{4}$ L
3. Ex 3: $\frac{1}{12}$
4. Ex 4: 1 goat
5. Ex 5: $\frac{4}{15}$
6. Ex 6: 64 pages

Set A

1. $\frac{5}{8}$
2. $\frac{1}{2}$
3. 24
4. $\frac{1}{3}$
5. $\frac{5}{8}$
6. 12

Set B

1. $2\frac{5}{6}$ m
2. $4\frac{1}{2}$ kg
3. $\frac{5}{12}$

Set C

1. $4\frac{7}{12}$ L
2. 1
3. 24 L
4. \$30
5. 12
6. 480 km

Set D

1. 100
2. $4\frac{9}{10}$ m
3. 63
4. No: they are equal (40)

Extension

1. 24
2. 300 mL

Exit ticket

1. Level 1: $\frac{11}{15}$
2. Level 2: 9
3. Level 3: \$70

4.09 Fractions on a calculator (booklet p. 35)**WE DO examples**

1. Ex 1: $\frac{11}{12}$
2. Ex 2: $1\frac{2}{5}$
3. Ex 3: $3\frac{2}{5}$; 3.4
4. Ex 4: $\frac{25}{72}$
5. Ex 5: $12\frac{1}{4}$
6. Ex 6: $5\frac{1}{16}$

Set A

1. $\frac{11}{12}$
2. $1\frac{19}{30}$
3. $\frac{19}{36}$
4. $\frac{25}{72}$
5. $4\frac{7}{12}$
6. $2\frac{5}{6}$

Set B

1. $\frac{1}{40}$
2. $1\frac{5}{12}$
3. $10\frac{8}{9}$
4. $2\frac{7}{24}$
5. 225
6. 28

Set C

1. $3\frac{2}{5}$
2. $5\frac{5}{8}$
3. $\frac{19}{8}$
4. 0.35
5. $\frac{5}{8}$
6. 0.3125

Set D

1. $12\frac{1}{4}$
2. $5\frac{9}{16}$
3. $2\frac{4}{15}$
4. $\approx 4 \times 2 = 8$; $8\frac{11}{80}$

Extension

1. $\frac{31}{32}$: it gets closer to 1
2. $\frac{13}{15}$

Exit ticket

1. Level 1: $\frac{31}{35}$
2. Level 2: $3\frac{1}{6}$
3. Level 3: 4

4.10 Percentages, fractions and decimals

(booklet p. 39)

WE DO examples

1. Ex 1: $\frac{13}{100}$; 0.13
2. Ex 2: $\frac{2}{25}$; 0.08
3. Ex 3: $\frac{3}{40}$; 0.075
4. Ex 4: 70%
5. Ex 5: 92%
6. Ex 6: 4.5%

Set A

1. $\frac{13}{100}$
2. $\frac{2}{25}$
3. $\frac{1}{2}$
4. $\frac{3}{4}$
5. $\frac{7}{20}$
6. $1\frac{1}{5}$

Set B

1. 0.13
2. 0.08
3. 0.45
4. 1.5
5. 0.025
6. 0.075

Set C

1. 70%
2. 92%
3. 75%
4. 60%
5. 4.5%
6. 130%

Set D

1. 0.375, 37.5%
2. $33\frac{1}{3}\%$
3. $\frac{1}{8}$
4. $\frac{3}{8}$, 38%, $0.4 = \frac{2}{5}$

Extension

1. $66\frac{2}{3}\%$
2. $\frac{37}{300}$

Exit ticket

1. Level 1: $\frac{1}{4}$
2. Level 2: 7%
3. Level 3: 18.75%

4.11 Percentage of a quantity (booklet p. 43)**WE DO examples**

1. Ex 1: 35
2. Ex 2: 21
3. Ex 3: \$9
4. Ex 4: 84
5. Ex 5: 60 kg
6. Ex 6: \$175

Set A

1. 35
2. 43
3. 21
4. 9
5. 9
6. 3

Set B

1. \$9
2. 84
3. 75
4. 27
5. 9
6. 60

Set C

1. 60 kg
2. \$63
3. 200 m
4. 81 min
5. 51
6. \$10

Set D

1. \$175
2. 9
3. \$46
4. Equal: both 21

Extension

1. \$57.60
2. No: 20% then 10% is 28% off

Exit ticket

1. Level 1: 8
2. Level 2: 70
3. Level 3: \$51

4.12 Expressing as a fraction or percentage (booklet p. 47)

WE DO examples

1. Ex 1: $\frac{17}{20}$; 85%
2. Ex 2: 75%
3. Ex 3: $\frac{1}{5}$
4. Ex 4: 20%
5. Ex 5: 37.5%
6. Ex 6: Mia 85%, Jake 84%: Mia

Set A

1. $\frac{3}{5}$
2. $\frac{3}{4}$
3. $\frac{17}{20}$
4. $\frac{2}{3}$
5. $\frac{2}{5}$
6. $\frac{1}{5}$

Set B

1. 70%
2. 75%
3. 85%
4. 65%
5. 90%
6. 37.5%

Set C

1. $\frac{1}{5}$, 20%
2. $\frac{3}{8}$, 37.5%
3. $\frac{1}{8}$, 12.5%
4. $\frac{2}{25}$, 8%
5. $\frac{3}{10}$, 30%
6. $\frac{1}{4}$, 25%

Set D

1. Mia (85% vs 84%)
2. 25%
3. 25%
4. 57.1%

Extension

1. 24%
2. 20%

Exit ticket

1. Level 1: $\frac{3}{4}$
2. Level 2: 84%
3. Level 3: 20%